

ACTIVITY 4

FOURIER SERIES/ FOURIER TRANSFORM

I. Objectives

1. To understand Fourier series/Fourier Transform representation of periodic signals.
2. To learn how to use MATLAB to find the Fourier Series/Fourier Transform of a discrete signal.
3. To identify the functions of Fourier series and Fourier transform and how they work.

II. Introductory Information

III. Data and Procedures

Approximating the Square wave Function Using Fourier Sine Series

Square Wave Function

The first function we examined which can be approximated by a Fourier series is the square wave function. This is a function which alternates between two function values periodically and instantaneously, as if the function was switched from on to off. The Square Wave function is also commonly called a step function. The function graphed from $x = -1$ to $x = 1$ is shown in Figure 1. By summing sine waves, it is possible to replicate the square wave function almost exactly, however, there is a discontinuity in this periodic function, meaning the Fourier series will also have a discontinuity. It is clear in Figure 1 that the discontinuity will appear at $x = 0$, where the functions jump from -1 to 1. The equation of this function is represented in Equation 1.

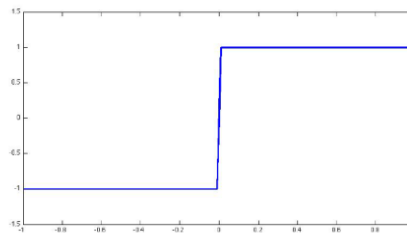


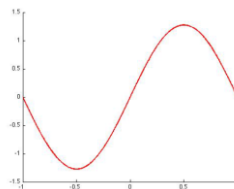
FIGURE 1: SQUARE WAVE FUNCTION

$$F(x) = \begin{cases} -1 & \text{for } -1 \leq x < 0 \\ 1 & \text{for } 0 \leq x \leq 1 \end{cases} \text{ eqn.1}$$

Fourier Sine Series

The Fourier sum of sines can be used to accurately approximate the square wave function. The more points plotted, and coefficients used the closer the Fourier sum will be to looking like the square wave function. The equation of Fourier sine series used in this case is represented in Equation 2. j represents the number of coefficients used. When using the sum of sines, only odd numbered values are used, otherwise you would be adding zero every other term. The starting point, where $j=1$ is shown in Figure 2.

$$F(x) = \frac{4}{\pi} \sum_{j=odd}^{\infty} \frac{1}{j \sin(j\pi x)}$$



Approximating with Fourier Sine Series

A MATLAB code is used to plot the square wave function along with the Fourier sine series to compare the accuracy and error between the approximation and the actual function.

Program 4_1

Matlab Code for Fourier Sum, Square Wave and Error

```
clf;
N = 1000;
x = linspace(-1,1,N);
f = sign(x);
sum = 0.*x;
M = input('Number of coefficient (M):');
for j = 1:2:M
    sum = sum + 4/pi*sin(j*pi*x)/j;
end
subplot(2,1,1);
plot(x, sum, 'r')
xlabel('Time index n');ylabel('Amplitude');
title('Fourier Series with M=1');
hold on
plot(x,f,'LineWidth',2)
hold on
subplot(2,1,2);
error = abs(sum-f);
plot(x,error);
xlabel('Time index n');ylabel('error');
title('Error with M=1');
```

Questions:

Q4.1 Run Program P4_1 and generate the Fourier series for square wave.

Q4.2 What will happen if the number of coefficient M is changed to 10, 50, 100, 1000 ? Discuss the result.

B. Fourier Transform

FFT algorithms are so commonly employed to compute DFTs that the term 'FFT' is often used to mean 'DFT' in colloquial settings. Formally, there is a clear distinction: 'DFT' refers to a mathematical transformation or function, regardless of how it is computed, whereas 'FFT' refers to a specific family of algorithms for computing DFTs.

The Discrete Fourier Transform of the sequence $x[n]$ can be defined as:

$$X[r] = \sum_{k=0}^{N-1} x[k] e^{-j2\pi kr/M} \quad (r = 0:N-1)$$

The inverse of it:

$$x[k] = \frac{1}{M} \sum_{r=0}^{M-1} X[r] e^{j2\pi \frac{kr}{M}}$$

Exercise: Calculate the four point DFT of aperiodic sequence $x[k]$ of length $N = 4$ which is defined by:



Program 4_2

Matlab Code for Discrete Fourier Transform

```
x = [2 3 -1 1];
N = length(x);
X = zeros(4,1)
for k = 0:N-1
    for n = 0:N-1
        X(k+1) = X(k+1) + x(n+1)*exp(-i*pi/2*n*k)
    end
end

t = 0:N-1
subplot(311)
stem(t,x);
xlabel('Time (s)');
ylabel('Amplitude');
title('Time domain - Input sequence')

subplot(312)
stem(t,X)
xlabel('Frequency');
ylabel('|X(k)|');
title('Frequency domain - Magnitude response')

subplot(313)
stem(t,angle(X))
xlabel('Frequency');
ylabel('Phase');
title('Frequency domain - Phase response')
```

X % to check $|X(k)|$
 angle(X) % to check phase

Questions:

Q4.3 Run Program P4_2 and generate the Discrete Fourier Transform.

Q4.4 Create a Matlab code that will calculate the five-point DFT of the aperiodic sequence $x[k]$ of length $N = 5$ which is defined as follows:

$$x[k] = \begin{cases} 3, & k = 0 \\ -1, & k = 1 \\ 1, & k = 2 \\ 4, & k = 3 \\ 2, & k = 4 \end{cases}$$

Q4.5 Create a Matlab code that will calculate the inverse DFT of

$$X[r] = \begin{cases} 5 & r = 0 \\ 3 - j2 & r = 1 \\ -3 & r = 2 \\ 3 + j2 & r = 3 \end{cases}$$

Fast Fourier Transform

The Fast Fourier Transform does not refer to a new or different type of Fourier transform. It refers to a very efficient algorithm for computing the DFT

Program 4_3

Matlab Code for Fast Fourier Transform

```
N=4;
x=[2 3 -1 1];

t=0:N-1;
subplot(311)
stem(t,x);
xlabel('Time (s)');
ylabel('Amplitude');
title('Input sequence')

subplot(312);
stem(0:N-1,abs(fft(x)));
xlabel('Frequency');
ylabel('|X(k)|');
title('Magnitude Response');

subplot(313);
stem(0:N-1,angle(fft(x)));
xlabel('Frequency');
ylabel('Phase');
title('Phase Response');
```

Q4.6 Run Program P4_3 and generate the Fast Fourier Transform.

Q4.7. Generate the Fast Fourier Transform using the given in Q4.4.

IV. Data and Results

V. Conclusion